

Logical Functions & Lookup

Question1. Write a formula to check if the marks in the cell are greater than or equal to 40. If yes, return , otherwise . B2 "Pass" "Fail".

Answer: Formula: **=IF(B2>=40,"Pass","Fail")**

Explanation: **B2>=40** → checks if marks are 40 or more

B2>=40 → checks if marks are 40 or more

"Fail" → shown if condition is FALSE

	A	B	C
1	Student Name,	Marks	Pass/Fail
2	Riya,45	45	Pass
3	Siya,72	72	Pass
4	Raju,55	55	Pass
5	Rahul,80	80	Pass
6	Ram,30	30	Fail
7			

Question2. A company gives a bonus if sales are above 40,000 and attendance is greater than or equal to 90%. Write a formula to calculate whether the employee will get a bonus or not .

Answer: Formula:

=IF(AND(B2>40000,C2>=90),"Bonus","No Bonus")

Explanation: **B2>40000** → Sales should be greater than 40,000

C2>=90 → Attendance should be 90% or more

AND → both conditions must be TRUE

"No Bonus" → if any condition fails

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Font

D3

=IF(AND(B3>40000,C3>=90),"Bonus","No Bonus")

Name Box

A

B

C

D

E

F

1

Employee

Sale

Attendance

Result

2

Arjun

55000

95

Bonus

3

Simran

65000

85

No Bonus

4

Rohit

30000

90

No Bonus

5

Anjali

25000

65

No Bonus

6

7

Question3. Write a nested IF formula: If marks in are $\geq 90 \rightarrow$, $\geq 75 \rightarrow$, $\geq 50 \rightarrow$, otherwise . B2 "A" "B" "C" "Fail".

Answer: Formula:

=IF(B2>=90,"A",IF(B2>=75,"B",IF(B2>=50,"C","Fail")))

Explanation: B2>=90 $\rightarrow$ Grade A

B2>=75 $\rightarrow$ Grade B

B2>=50 $\rightarrow$ Grade C

Otherwise $\rightarrow$ Fail

Question4. Find the Department of Employee ID = E103. Also explain which LOOKUP function is used and why?

Answer: use **VLOOKUP**

Reason: 1. The data is arranged in vertical columns

2. The lookup value (**EmpID**) is in the **first column**

3. We want to return a value (**Department**) from a column on the **right side**

-> Therefore, **VLOOKUP** is the correct function for this case

=VLOOKUP("E103",A2:C5,3,FALSE)

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F
1	EmpID	Name	Department			
2	E101	Raju	IT			
3	E102	Riya	HR			
4	E103	Roshni	Finance			
5	E104	Sejal	Sales			
6						
7						
8						
9		Department				
10		=VLOOKUP("E103",A2:C5,3,FALSE)				
11						
12						

The formula bar at the top shows: **=VLOOKUP("E103",A2:C5,3,FALSE)**

A tooltip for the VLOOKUP function is visible at the bottom, showing the syntax: **VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])**

Question5: company has set the following commission structure for its sales team: If a salesperson achieves sales of ₹45,000, use

the VLOOKUP function to determine the commission rate applicable.

Answer: Sales = 45,000 , Applicable Commission Rate = 10%

Formula (VLOOKUP – Approximate Match)

=VLOOKUP(45000,A2:B5,2,TRUE)

Explanation: -> The table is arranged in ascending order of sales amount.

→ We use VLOOKUP with approximate match (TRUE)

→ When the exact value is not found, VLOOKUP returns the nearest smaller value.

Here:

→ 45,000 is greater than 20,000 but less than 50,000

→ So, it picks the rate for 20,000, which is 10%

	A	B	C	D
1	Sales Amount	Commission Rate	Commission Rate	
2	0	5%	0.1	
3	20000	10%		
4	50000	15%		
5	80000	20%		
6				
7				

Question6.The HR department maintains the following leave records of employees:onus or not .

Write an IF function to check whether an employee has Exceeded their allowed leave balance. If Leaves Taken > Allowed Leaves, display "Exceeded". Otherwise, display "OK".

Answer: Formula: **=IF(B2>C2,"Exceeded","OK")**

Explanation:

→ B2 = Leaves Taken

→ C2 = Leaves Allowed

→ Formula checks condition:

→ If $B2 > C2 \rightarrow \text{"Exceeded"}$

Else → "OK"

In this case, no one exceeded their leave limit, so all results are "OK".

D2

✖

✓

fx

=IF(B2>C2,"Exceeded","OK")

	A	B	C	D	E
1	Employee Name	Leaves Taken	Leaves Allowed		
2	Alex	1000%	12	OK	
3	Bob	1000%	15		
4	Anna	800%	12		

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✖

Question7 : Find the salary of EmpID = E105 using INDEX-MATCH Function.

Answer: Formula: `=INDEX(B2:B5, MATCH("E105", A2:A5, 0))`

Result:

E105 ka Salary = 72000

Explanation: → A2:A5 = EmplID column

→ B2:B5 = Salary column

→ MATCH("E105", A2:A5, 0) → E105 ki position find karta hai (Result = 2)

→ INDEX(B2:B5, 2) → Salary column ka 2nd value return karta hai

→ Final result = 72000

Formula Bar: `=INDEX(B2:B5, MATCH("E105", A2:A5, 0))`

	A	B	C
1	EmpID	Salary	
2	E104	₹ 55,000.00	72000
3	E105	₹ 72,000.00	
4	E106	₹ 62,000.00	
5	E102	₹ 41,000.00	
6			

Question 8. Use the XLOOKUP function to find the Department of the employee with Emp ID = E303.

Answer: Formula: **=XLOOKUP("E303", A2:A5, B2:B5)**

Explanation: → "E303" = jis value ko search karna hai

→ A2:A5 = EmpID column (lookup range)

→ B2:B5 = Department column (return range)

→ XLOOKUP E303 ko dhundta hai aur uska department return karta hai

→ Final result = IT

Result: IT

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C2			=XLOOKUP("E303", A2:A5, B2:B5)		
	A	B	C		
1	EmpID	Department			
2	E101	HR	IT		
3	E303	IT			
4	E106	Sales			
5	E301	Finance			